

Exp 10 Digital Speech Processing and Noise Reduction

10.1 Implementation of Spectral Subtraction for Speech Noise Reduction

Program:

```
clc; clear; close all;

Fs=8000; t=0:1/Fs:3;

x=sin(2*pi*200*t)+0.5*sin(2*pi*400*t); % Clean signal

xn=x+0.3*randn(size(x));           % Noisy signal

Y=fft(xn);

N=fft(0.3*randn(size(x)));

xe=real(ifft(max(abs(Y)-abs(N),0).*exp(1j*angle(Y))));

% Signals

figure;

subplot(311),plot(t,x),title('Clean Speech')

subplot(312),plot(t,xn),title('Noisy Speech')

subplot(313),plot(t,xe),title('Enhanced Speech')

% Spectrograms

figure;

subplot(311),spectrogram(x),title('Clean Spectrogram')

subplot(312),spectrogram(xn),title('Noisy Spectrogram')

subplot(313),spectrogram(xe),title('Enhanced Spectrogram')

% Magnitude Spectra

figure;

subplot(311),plot(abs(fft(x))),title('Clean Spectrum')

subplot(312),plot(abs(fft(xn))),title('Noisy Spectrum')

subplot(313),plot(abs(fft(xe))),title('Enhanced Spectrum')
```

Outputs:

